

Top Insights from Chat

Questions · Answers · Fixes the Cohort Found Together

The chat during Session 8 was unusually good. Alongside the expected setup questions, the cohort produced a steady stream of sharp architectural questions, several of which the mentor pulled out and answered live, and a few learner answers that were better than anything a slide could have delivered. What follows is the substance, grouped by theme, with every question attributed to the person who asked it.

1. Choosing Between Single and Multi-Agent

The most valuable thread of the session. Almost everyone wanted the same thing: a rule for deciding when the extra machinery is worth it.

Asked by	Question	Answer or takeaway
Sanjeev Rohila	Any scenario from you (examples) where you will choose single or multi-agent?	Deferred to the end of the session deliberately, so the class could work out examples themselves first. The four-criteria decision table (complexity, latency, cost, accuracy) is the answer.
sibasis	I am automating a GCP load process of 8 steps, Excel comparison and DB queries. My manager says go multi-agent, I think a single agent will work. How do I prove it?	If the order is fixed or sequential, a sequential workflow is better. Move to multi-agent only when you need dynamic routing or you are hitting context limits. You prove it by comparison, experimentation and evaluation.
sibasis	How do you know if the pattern you chose is correct or not?	Comparison, experimentation and evaluation. Always start with the simplest approach. If it does not meet expectations under evaluation, gradually move to a more complex architecture. Never start complex and try to simplify later.
Goutham	How do you decide what architecture to go with for what use case?	Map the business problem to a workflow first. See which isolated pieces or specialised tasks it breaks into. Distinguish agents by their specialisation, the same way you would scope roles in a real organisation.

Asked by	Question	Answer or takeaway
Sharath S	Is there a mental model for deciding the scope of agents in a multi-agent setup? I am drawing parallels to microservice architecture and DDD.	The parallel holds. You decide the boundary of an agent the way an organisation decides where a product manager's job ends and a programmer's begins: by specialisation.
Sharath S	When do we define the workflow ourselves, and when do we let the agent create the workflow dynamically? What is the trade-off?	Define it yourself when you have a clear idea of the workflow and most steps are static. Make it agentic when decisions have to be made while the workflow is running, so the LLM can decide on the go.
Pujan	Can we know when to use RAG, when to use a single agent, and when to use a multi-agent system?	Same decision framework. Start simple, evaluate, and escalate only when the simpler design demonstrably fails.

Key Insight (from shanmukhh vadrevu)

"The excessive supervisory layers create unnecessary latency, token costs, and communication bottlenecks for a simple workflow." This was posted in response to the deliberately over-engineered research example, and it is the cleanest one-line statement of the overkill problem anyone gave.

2. How the Architectures Actually Behave

Asked by	Question	Answer or takeaway
Pujan	Will there be something like a Router, and how do we decide when to choose which?	You can use one, but be clear on the two use cases. A router chooses the path. A multi-agent system coordinates the workers.
anuragp	Does the coordinator agent have access to tools as well, or do only the sub-agents have tools?	It may or may not have direct tool access. That depends entirely on the use case.
anuragp	Will each sub-agent still be a ReAct agent?	Yes. A ReAct loop is running inside every one of those blocks.
Pawan	Will they run in parallel?	They can, if you design them to. Usually they do not, because one agent's output often depends on another's. Run them in parallel only when there is no dependency.
Amit Patil		

Asked by	Question	Answer or takeaway
	Supervisor and hierarchical involve running agents in parallel, correct me if I am wrong?	Both can, but it is entirely your choice. It is not inherent to either pattern.
Sandip	How does the Chinese whispers example fit hierarchical, when the sub-agents are not talking to each other?	The corruption travels through the manager. If one agent is context poisoned and returns an irrelevant response, that message lives in the manager's history and trickles down into the other sub-agents. Sandip's own name for it: manager's mess.
Renganathan M	The sequential pattern should have low latency since the final result arrives after the final node runs, whereas hierarchical is a pattern. Am I right?	Correct on sequential. In hierarchical it depends on when the supervisor node is satisfied with the responses of all sub-agents, and the communication itself costs time and compute, which is why it is both the most expensive and the longest running.
AMEY BARBADIKAR	Can the end node be assigned to more than one agent?	It depends. There are also web-like architectures where each agent can communicate with every other. The three patterns taught are the basic ones, not the only ones.
RK	How do we decide which agent runs after which one? Do we decide it using LangGraph and our use case?	Through edges, and conditional edges for routing. Each agent is a node, the edges decide what runs next. The chat answered this one before the mentor did.
anuragp	When some nodes become agentic, how is the edge flow respected? Do we decide beforehand, or does the node agent decide which edge to follow?	The edges are still defined beforehand in LangGraph. An agentic node runs its own internal ReAct loop, but once it finishes, LangGraph follows the edge you defined.
Amit Patil	Can each subgraph or agent in a hierarchical architecture be reused for different use cases and architectures?	Yes, that can be done, depending on the use case.
Jagadeesh	What type of architecture does VS Code Copilot or Claude Code use?	Systems like Claude Code, OpenClaw and Hermes use the process of creating sub-agents on the fly. That is the hierarchical family.
Pujan	Was that real scenario, where an agent was creating an agent by itself, part of hierarchical?	Yes. Claude Code and OpenClaw are the live examples of it.

3. Reliability, Loops and Failure Modes

Asked by	Question	Answer or takeaway
Yasmin Iqbal	Is it possible for an agent to get locked in an infinite loop?	Very possible, which is why you set max iterations or retries. The mentor's own agent ran for around sixteen minutes on Groq because of a middleware bug.
Renganathan M	What was the mistake? Just curious.	He used a summariser agent to minimise context propagation, but it was trimming rather than summarising. The meat of each tool's output was being cut, so the agent concluded the tool had failed and kept retrying.
Ravindra Devagiri	Is it possible to know whether an agent is stuck in an infinite loop, or stuck forever for some other reason?	Detect it with max iterations and timeouts.
Annu	How do we ensure accuracy and reliability of end-user responses in a multi-agent system, particularly when responses are generated, transformed or rephrased by sub-agents, while minimising hallucination risk?	Enforce structured outputs using Pydantic or JSON, and validate before passing results from one agent to the next. Expanded in the production nuggets: validation gates, retry caps, backoff, and an eval layer.
Sandip	Some people got a PR and some did not with the same model. It becomes probabilistic. Somewhere the agent is not catching something.	Exactly right, and the mentor confirmed the mechanism. Learners who got a PR without changing the code got it because their model worked out that the code review agent had PR access and routed there. Same code, different route.
Murthy Rajender	Rerun as many times as you like, it should give the same output.	It does not, and that is the point. Stochastic routing is a property of the system you have to engineer around, not a bug to be surprised by.
anuragp	Can we get better results with lower models but using hierarchical task decomposition?	That was the whole idea behind it. Cheap sub-agents plus a moderate supervisor can perform a complex task at lower cost, and most of the time it works, but only if the context shared between the nodes is really well defined.
Goutham	Can a whole complex workflow rely on only one model? Will there be cases	Yes to both, and this is the mechanism behind the previous answer. Mixed model tiers across

Asked by	Question	Answer or takeaway
	where a sub-agent needs an extra LLM of its own besides tools and memory?	the hierarchy is how you trade cost against capability.

4. Understanding the Code

Asked by	Question	Answer or takeaway
Renganathan M and Karthikeyan M S	Just by using the word "supervisor", we created the supervisor agent? I do not see any specific way of establishing the relationship between supervisor and sub-agents.	The relationship is established by the tools list. Syed Mizbahuddin answered it in the chat with the exact code: <code>github_agent = create_agent(model=llm, tools=[call_documentation_agent, call_release_notes_agent, call_issue_agent, call_code_review_agent])</code> . The GitHub agent is the supervisor, and you are giving it access to the other agents.
Sharath S	Is it always required to convert sub-agents to tools? Can we not give sub-agents to a supervisor agent without the tool wrapper?	You can, but you would need a middleware from the Deep Agents framework, where sub-agents can be passed as a Python dictionary or JSON with name, system prompt and description. The tool wrapper is the simplest way with what has been taught so far.
Karthik Prabhu	Can you share how you created the system prompts defined in each of the sub-agents?	Give the agent a role first, then the task, then the actions it can perform. The named method is CRAFT: Context, Role, Action, Format, Tone.
Aditya Bharadwaj	How do you get into the mindset of what state and what nodes you need when designing systems with LangGraph? I am especially curious about states.	The state is the data that must persist across the workflow. Then define each action as a node, and use edges to decide the next node based on the state.
Renganathan M	If the API names are good enough, do we not need the tool name mapping?	Correct. The mapping exists because the GitHub toolkit ships human readable names with spaces, and agents treat tools as functions. If your names are already function-like, skip it.
Abhisekh	The tool names are from GitHub?	Yes. They are real GitHub tools with human readable names, converted to AI readable names. Nothing else changed.

Asked by	Question	Answer or takeaway
srinivas	If this tool acts as an agent, then shouldn't the response be an AIMessage?	No, it must be a tool message, because the sub-agent is acting as a tool for the supervisor agent. That is what the Command wraps and returns.
Jagadeesh	Can you explain Command more?	Command is how you do agent to agent communication in LangChain and LangGraph, and more generally how you inject a message into a pipeline from a tool, a middleware or an agent. The cohort had already used it in the human in the loop session.
Amit Patil	How does it know the sequence? Or is it decided by the model?	The model decides. The sequence was never programmed. The supervisor decides on the go, which is what makes it agentic.
anuragp	Can creating sub-agents also be done on the fly by the supervisor?	Not here. The sub-agents are defined ahead of time and the supervisor picks between them. It is not creating agents on the fly.
Pujan	Do we not need human in the loop here?	It is already there in a different flavour. The pull request is the human checkpoint. The agent cannot write to main directly.
Karthik Prabhu	Doesn't it make more sense to introduce human in the loop before generating the PR?	Human in the loop comes in various flavours, and the PR itself is one of them.
renjith r	Can these agents invoke simple statistical ML models in addition to an LLM?	Yes. Syed Mizbahuddin answered it well in the chat: a tool can do anything, it is just a Python function.
Ashu Swami	How did other people get the pull request without adding the tool? I did not add it and still got a PR.	Their model worked out that if the documentation agent lacked the PR tool, the code review agent had it, and routed there. Stochastic routing.
Aditya Bharadwaj	How did you figure out that you had to add PR to the definition?	By reading the agent's trace. The trace showed it never had the tool it needed to perform the action.
Karthik Prabhu	Is there any way to debug how agents do the thinking before taking action?	Yes, and it is exactly what the mentor did live. Observability gets its own treatment in Session 9.

Key Insight (from chaithanya krishna)

"Tools are just functions with access to models, and the whole system can be considered as function calling. The framework of the workflow is available with LangGraph, and the way we use it determines the sequential, supervisor or hierarchical pattern." The mentor's response: you have very well understood whatever we have learned so far.

5. Setup and Environment: The Fixes That Worked

A large part of the hands-on was spent on setup, and the cohort solved most of it for each other in the chat. These are the fixes worth keeping, because you will hit them again.

Symptom	Fix, and who supplied it
uv is not recognized as a cmdlet	uv is not installed. Install it from the official uv installation docs for your OS rather than via pip, so the environment is not tied to a particular Python or pip install. (Ck Grandhi hit it, Sid pointed to the docs.)
No pyproject.toml found in current directory or any parent	You are in the wrong folder. Change into the cloned github-agent repo and run from there. (Kaviarasu RC, Venkateswara Rao K, sibasis; answered by Dhaval Patel and Amit Patil.)
No Scripts folder when activating the venv	That is Windows only. On macOS and Linux use <code>source .venv/bin/activate</code> . Sayee Sree Ram posted the full table of activation commands per shell, which the whole team endorsed.
ImportError: attempted relative import with no known parent package	Run it as a module: <code>uv run python -m src.agents.graph</code> . This was the single most repeated fix of the session, first posted by Syed Mizbahuddin.
jwt.exceptions.InvalidKeyError: Could not parse the provided public key	The .pem path in GITHUB_APP_PRIVATE_KEY is wrong, the file was edited or renamed, or the slashes are wrong for Windows. Paste the full absolute path and check the file is intact. (Pujan, Pradeep Deshmukh, sanjeev rohila; fix from Syed Mizbahuddin.)
GithubException UnknownObjectException: 404 Not Found	GITHUB_REPOSITORY must be <code>username/repository-name</code> , not just the repo name and not a full https URL. Also confirm the GitHub App is installed on that specific repo. (Shraddha Mishra, Venkat A; fixes from Syed Mizbahuddin and SHIBA.)
DeprecationWarning: langchain-community is being sunset	A warning, not an error. Ignore it. The migration off langchain-community is a planned future session.

Symptom	Fix, and who supplied it
groq RateLimitError 429, tokens per minute exceeded	Free tier limit. Retry, it works on a later attempt. (Aditi Reddy hit it and confirmed retrying worked.)
groq PermissionDeniedError 403, check your network settings	Srikanth Adilapuram resolved it by disconnecting his VPN.
Corporate network blocking files.pythonhosted.org	An IT admin block. Use a personal machine and a personal network or hotspot. (Ankit Garg.)
Agent ran with no error but nothing changed in the repo	Check the branches, not main. The agent creates a feature branch. (Yasmin Iqbal; answered by Syed Mizbahuddin.)
Danger zone appears after installing the app	Not an error. It only means those options can uninstall or suspend the app. (Raised by several, including Shruti Penumetsa and Bharath Kumar.)

Key Insight (from Ashish, Team Codebasics)

A `.env` file contains private and confidential credentials, which is why the project ships a `sample.env` with placeholders instead. Create your own `.env` and replace the placeholders with your own keys. Never share your `.env`, never expose your API keys publicly, never commit them, and never paste them into chat. Sandip's addition: put both `.pem` and `.env` in `.gitignore`.

6. Standout Participant Contributions

Several people answered questions the mentor had not got to yet, or added something the session was better for. Worth calling out.

From	Contribution
sunil	Posted the full CRAFT breakdown (Context, Role, Action, Format, Tone or Tweaks) when the mentor named the acronym, and it was adopted straight into the teaching. He also correctly guessed the agent-as-tool wrapper before it was explained.
Syed Mizbahuddin	The most active answerer in the room by a wide margin. Supplied the correct run command, the supervisor code snippet, the 404 and JWT fixes, and a fully worked multi-agent bug-solver design early in the session.
Sayee Sree Ram	Posted the complete virtual environment activation table across Windows CMD, PowerShell, Git Bash, macOS and Linux, which unblocked a large part of the room at once.
Ravindra Devagiri	

From	Contribution
	Compressed the entire code walkthrough into three lines: "Call a task as function, function as agent, and agent as tool." Also posted a working pyproject.toml for people who could not find theirs.
shanmukhh vadrevu	Wrote out a five-step design for a ChatGroq class with automatic fallback to other free Groq models on rate limit, for everyone hitting the free tier ceiling.
Renganathan M	Named the sequential pattern rule before it was taught: if the input of one agent depends on the output of the previous one, that is the sequential pattern.
anuragp	Identified OpenClaw as the hierarchical example the mentor was fishing for, and named separation of concerns as a benefit of multi-agent systems, leading to less hallucination.
Reshma	Answered a peer's question about outcomes from her own experience: she got her job after completing the Codebasics Data Analyst course three years ago, said it helped her prepare for interviews and gave her projects to show, and is now building AI skills.
komal	Named the most transferable skill of the day: learning how to use the Antigravity chat to debug was the most useful part for her. The mentor agreed.

7. Tools and Environments the Cohort Is Using

Tool	What the room said
Antigravity	The mentor's IDE for the session, alongside Claude Code. Bharath Kumar: awesome, especially for debugging. Sid on the Copilot comparison: Copilot is integrated by default in VS Code, but the free limit is not generous and Antigravity is better.
Cursor	Yasmin Iqbal: very easy to use, but very limited tokens. Tony Onabanjo asked for the comparison with Antigravity; the mentor had not used Cursor.
uv	Sandip's summary: a package manager that resolves dependencies and is faster than pip. Krishna: uv sync installs all dependencies and creates the venv.
Groq free tier	Rate limits were the single most common runtime failure. Retry, or implement a fallback across free models.
CrewAI	Named by the mentor as the well known framework that follows a hierarchical structure by default.

8. Feedback on the Session

The mentor asked directly whether the pace was too fast, and the room answered honestly. Worth recording, because the answers were specific rather than polite.

From	Feedback
Rakkappan	Today's session was too fast.
Bharath Kumar	Not fast, but overwhelming. Needs a week to review, digest and assimilate. Asked whether others felt the same.
Renganathan M	The session was great, but so condensed that splitting it into two sessions would have helped connect the dots better.
Sharath S and umesh	Setup took time and that part was not productive.
Tony Onabanjo	Pace was perfect.
RK	Good, with fewer distractions from chat questions. Reading questions out loud before answering, and batching them into designated slots, helped.
Pujan	Good hands-on session, but needs a revision pass over all of the agent material including LangChain and LangGraph.
anuragp	Best is to revisit the lectures.

Key Insight (the mentor's reply to Bharath Kumar)

It is okay to feel overwhelmed. You have the whole week. If you revise it a few times it will start to seem easy. All you need is revision and practice.